

Stochastic Dynamic Games in Belief Space

Autonomous Navigation & Perception (086762) Spring 2025 - Project

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1 Introduction

Planning for **multi-agent systems** is one of the stickiest problems in modern robotics. Think of fleets of self-driving cars, warehouse robots weaving around each other, or drones sharing crowded airspace: every robot has its own goal, yet each one’s decision making alter what the others will face next. Add the fact that sensors are noisy and actions don’t always have the intended effect, and suddenly every decision is made on shaky ground.

The usual fix for uncertainty is to plan in *belief space*—that is, over probability distributions rather than single “true” states—using tools from *Partially Observable Markov Decision Processes (POMDPs)*. POMDPs work great when a lone robot needs to be cautious, but they don’t capture strategic give-and-take between multiple robots. Classic Game Theory, on the other hand, nails the strategic side yet normally assumes everyone sees the world perfectly—hardly realistic.

The paper “*Stochastic Dynamic Games in Belief Space*” [1] **stitches these two views together**. It frames multi-robot interaction as a *stochastic dynamic (differential) game in Gaussian belief space* and solves it with a belief-space version of iterative LQG (iLQG). Each agent reasons about the probability distributions of everyone’s states (its own included) *and* about how its moves will trigger responses from the others, which in general govern the definition behind *first order beliefs*. The output is a feedback policy that stays robust to sensor noise while remaining tactically smart.

How is this relevant for our **Autonomous Navigation and Perception (086762)** course ? Because it covers our mains course topics —POMDPs, optimal control, and BSP in one framework. It suggests a pseudo-code algorithm for corresponding multi-sgent online planning problems, and does so with a runtime that scales linearly in the planning horizon and only polynomially in the number of agents. That’s a big leap compared with the usual exponential blow-ups we fight in multi-agent POMDPs.

Schwarting *et al.* backs their theory with well constructed and based case-studies:

- **Autonomous racing:** an ego-car chooses lines that unsettle rivals while managing its own uncertainty.
- **Surveillance / herding:** a drone gently “pushes” targets toward well-lit areas to sharpen its estimates.
- **Assistive guidance:** a helper robot selects paths that

both achieve the task and reduce a human partner’s uncertainty while being blind (observation less).

In short, the paper shows how multiple autonomous agents can act *strategically, robustly, and without explicit chatter..* This report will unpack the mathematics behind that claim, weigh the strengths and weaknesses of the approach, suggest where the idea could go next, and provide a parallel simulation improving and investigating what was proposed.

Key Concepts Primer

Belief-Space Planning (BSP). Extends stochastic optimal control to the *space of probability distributions*, allowing an agent to value information gain as well as classical performance costs.

POMDP viewpoint. A BSP problem can be written as a partially observable Markov decision process tuple $\langle \mathcal{X}, \mathcal{A}, \mathcal{Z}, b_0, P_T, P_Z, C \rangle$ whose exact solution is NP-hard; approximations are therefore mandatory.

Game-theoretic layer. When $N > 1$ self-interested agents interact, we seek a (local) Nash equilibrium in *belief space* which will reflect the realistic cognitive reasoning of the agents, rather than a single-agent optimum.

iLQG in belief space. By linearising the belief dynamics and quadratising the costs expressions, the authors obtain a backward-forward *belief-space iLQG* whose feedback gains K_k couple all agents.

First-order beliefs. The algorithm reasons up to “I think they think...” only; governed by *the cognitive theory of the mind*. Higher-order belief hierarchies explode combinatorially and are left for future work [1].

Nash Equilibrium. A Nash equilibrium is a set of strategies $\{u_k^{(i)}\}_{i=1}^N$ such that no agent can unilaterally reduce its own cost by deviating from its strategy, assuming the others keep theirs fixed:

$$Q^{(i)}(\mathbf{b}_0, u_{0:T-1}^{(i)}, u_{0:T-1}^{(-i)}) \leq Q^{(i)}(\mathbf{b}_0, \pi_{0:T-1}^{(i)}, u_{0:T-1}^{(-i)})$$

$$\forall i \in \{1, \dots, N\}.$$

where Q is the action-value function, measuring the long term influence on the cost chosen, while maintaining in this case the initial belief and the controls of the other agents in the environment, and π is the optimal policy at a certain time instant. In belief-space games, each agent’s cost depends not only on its own actions but also on its estimate of other agents’ future actions and beliefs.

2 Related Work

2.1 Prior and Contemporary Work

The contribution of Schwarting et al. [1] lies at the intersection of three major lines of research: belief space planning (BSP), game-theoretic planning, and local trajectory optimization methods such as iLQG. Each of these areas has been extensively studied in the robotics and decision-making communities, but their integration in a unified framework remains rare, and in paper it was even mentioned that “this is the first work to combine general dynamic games and planning in belief space into an efficient real time algorithm”.

Lets walk through these 3 major lines.

Belief Space Planning (BSP) BSP addresses the problem of planning under uncertainty by considering not the system’s state directly, but its belief over the state, typically modeled as a probability distribution. This framework is formally grounded in the theory of Partially Observable Markov Decision Processes (POMDPs) [2]. Solving generic POMDPs optimally is computationally intractable, which has led to the development of approximate methods. Among the most influential are point-based algorithms such as SARSOP [3], which restrict the planning to reachable belief subsets. However, these methods are primarily designed for discrete state spaces and scale poorly with planning horizon length.

In contrast, optimization-based BSP approaches in continuous domains have gained traction. Van den Berg et al. [4], Patil et al. [5], and Sun et al. [6] proposed techniques that approximate beliefs as Gaussians and solve continuous POMDPs locally using trajectory optimization. These methods achieve scalability by linearizing belief dynamics and quadratizing cost functions, but are typically limited to single-agent scenarios. Schwarting et al.’s work builds directly upon this foundation, extending local BSP to multi-agent interactive settings.

Game-Theoretic Planning and Multi-Agent Interaction In parallel, significant work has explored multi-agent interactions using game theory. Many of these studies model human-robot interaction, autonomous driving, or multi-robot collaboration through noncooperative dynamic games. For instance, Sadigh et al. [7] and Schwarting et al. [8] employed best-response models to plan in human-centric environments, leverag-

ing inverse reinforcement learning (IRL) to infer human preferences. Williams et al. [9] and Spica et al. [10] studied competitive interactions such as autonomous racing, where agents must predict and react to opponents’ strategies.

These approaches generally compute Nash equilibria or best-response strategies **under full observability** or simplistic noise models. Although game-theoretic reasoning enables agents to anticipate others’ reactions, it often omits detailed uncertainty modeling—particularly when it comes to belief evolution over time. The work of Schwarting et al. addresses this gap by embedding game-theoretic reasoning directly in belief space, allowing agents to optimize while accounting for both strategic interaction and uncertainty.

iLQG and Local Trajectory Optimization under Uncertainty Iterative Linear Quadratic Gaussian control (iLQG) [11] is a widely used method for solving stochastic optimal control problems by approximating nonlinear dynamics and cost functions around a nominal trajectory. Extensions of iLQG to belief space, such as those by Van den Berg et al. [4] and Sun et al. [6], allow planning under uncertainty by propagating Gaussian beliefs with an extended Kalman filter. However, these formulations assume a single agent and ignore interactions.

In parallel, solutions to multi-agent games such as ALGAMES [12] or iterated best-response schemes [13] compute Nash equilibria using local approximations, but typically in state space, not belief space. The novelty of Schwarting et al.’s approach is in solving a quadratic game at each step of the iLQG backward pass within belief space. This enables efficient planning in continuous, partially observable, multi-agent environments, achieving runtime complexity linear in the planning horizon and polynomial in the number of agents.

In summary, while prior work laid the theoretical and algorithmic foundations for either belief-space planning or dynamic games, none provided a scalable solution to dynamic multi-agent decision-making under uncertainty. Schwarting et al.’s contribution consists precisely in integrating these paradigms—BSP, game-theoretic reasoning, and local trajectory optimization—into a unified, tractable algorithm.

2.2 Subsequent Developments Since Publication

Since its publication, the work of Schwarting et al. [1] has attracted attention for its integration of belief-space planning and dynamic game theory.

By the time the paper was written a work that cited Schwarting et al. [1] was *Potential iLQR: A Potential-Minimizing Controller for Planning Multi-Agent Interactive Trajectories*, by Talha Kavuncu and Ayberk Yaraneri and Negar Mehr [14], which proposed a specialized formulation for multi-agent planning by leveraging the structure of potential games. Unlike the iterative best-response approach in [1], which requires separate backward passes **for each agent**, the potential cost minimizing method enables **a single collective optimization** over a shared potential function. This shift greatly reduces computational complexity while retaining the ability to model strategic interactions, making it particularly attractive for real-time multi-agent trajectory generation under uncertainty. The work can thus be seen as a natural progression from the general stochastic dynamic game framework of [1] to a more efficient but structurally constrained optimization scheme.

One significant direction of follow-up research addresses the challenge of computational scalability in multi-agent settings, which remains a practical limitation for the broader deployment of the original framework. A representative example is **the recent work** by Vakil, Coffey, and Pierson [15], who propose a method called *Partial Belief Space Planning* to scale stochastic dynamic games. Their contribution focuses on reducing the dimensionality of belief propagation within the iLQG-based planner. Instead of maintaining and updating the full Gaussian belief space over all agent states—which becomes computationally expensive in large systems—they selectively update a subset of belief components deemed most relevant for strategic planning. This approximation allows for near real-time execution while preserving the critical structure of the belief-space Nash equilibrium computation.

Importantly, their method retains the original game-theoretic feedback planning structure introduced in [1], including the use of belief-space dynamics, Nash equilibrium computation in the backward pass, and local trajectory optimization. Their experiments demonstrate that partial belief propagation achieves competitive performance with significantly lower computational

cost in surveillance and racing scenarios, validating its utility in larger-scale multi-agent systems.

This contribution illustrates the methodological flexibility of the belief-space game-theoretic framework and its potential for adaptation to more scalable and efficient implementations.

3 Preliminary material and Problem formulation

This section establishes the notation, mathematical background, and formal problem statement on which the rest of the report builds.

3.1 Notation and Symbols

We adopt a discrete time index $k \in \{0, \dots, T\}$. Bold symbols denote joint (multi-agent) quantities, while plain symbols refer to a single agent $i \in \mathcal{A}$. A compact symbol table is provided for quick reference in [Appendix](#).

3.2 System and Observation Models

Each agent obeys nonlinear stochastic dynamics and noisy sensing:

$$x_{k+1}^i = f(x_k^i, u_k^i) + M_k^i w_k^i, \quad w_k^i \sim \mathcal{N}(0, I), \quad (1)$$

$$z_k^i = h(x_k^i) + N_k^i v_k^i, \quad v_k^i \sim \mathcal{N}(0, I). \quad (2)$$

Intuition. Equation (1) says the next state equals a deterministic motion model $f(\cdot)$ plus Gaussian process noise scaled by M_k^i , while (2) expresses the sensor reading as the true state passed through $h(\cdot)$ plus measurement noise. The functions f and h are assumed C^1 ; their linearisation about a nominal trajectory is detailed in Section 4.

3.3 Mathematical Practical Representation

In paper, he suggested a practical implementation of the Bayesian filtering step that underpins belief-space planning. The theoretical form,

$$\mathbf{b}_{k+1} = \psi(\mathbf{b}_k, \mathbf{u}_k, \mathbf{z}_{k+1}),$$

expresses the Bayesian update of the belief $\mathbf{b}$ using the prior belief $\mathbf{b}_k$, the applied control $\mathbf{u}_k$, and the received observation $\mathbf{z}_{k+1}$. In practice, the paper reformulates this update for computational

efficiency by decomposing it into a deterministic part, $g(\mathbf{b}_k, \mathbf{u}_k)$, and a stochastic part, $W(\mathbf{b}_k, \mathbf{u}_k) \xi_k$, where $\xi_k \sim \mathcal{N}(0, I)$. The deterministic component propagates the mean state and reshaped covariance vector, while the stochastic component injects Gaussian process noise into the belief dynamics:

$$\hat{\mathbf{b}}_{k+1} = \underbrace{g(\mathbf{b}_k, \mathbf{u}_k)}_{\text{Deterministic}} + \underbrace{W(\mathbf{b}_k, \mathbf{u}_k) \xi_k}_{\text{Stochastic}}, \quad \xi_k \sim \mathcal{N}(0, I),$$

with

$$g_k(\mathbf{b}_k, \mathbf{u}_k) = \begin{bmatrix} f(\hat{\mathbf{x}}_k, \mathbf{u}_k, 0) \\ \text{vec}(\Gamma_{k+1} - K_k H_k \Gamma_{k+1}) \end{bmatrix},$$

$$W_k(\mathbf{b}_k, \mathbf{u}_k) = \begin{bmatrix} \sqrt{K_k H_k \Gamma_{k+1}} \\ 0 \end{bmatrix}.$$

Here, f denotes the state transition model under zero noise, Γ_{k+1} is the predicted covariance, K_k is the Kalman gain, and H_k the observation matrix. This formulation is central to the paper's belief-space game-theoretic planning because it enables the propagation of uncertainty in a manner that is both analytically tractable and efficient for optimization within the iLQG framework.

Value Function Theoretically, the value function of agent i in a stochastic dynamic game over the belief state $\mathbf{b}_k$ is defined as

$$V_k^i(\mathbf{b}_k) \triangleq Q_k^i(\mathbf{b}_k, \pi_k) \triangleq \mathbb{E}_{\mathbf{z}} \left[c_l^i(\mathbf{b}_l) + \sum_{k=0}^{l-1} c_k^i(\mathbf{b}_k, \pi_k) \right] \quad (3)$$

In practice, it is not possible to calculate such value function according to the definition, because it means that we need to consider all possible observations at each time instant. In practice, the value function $V_k^i(\mathbf{b}_k)$, is computed using a quadratic approximation, consistent with iLQG usage, to enable efficient backward propagation. According to *Corollary 2*, this leads to discrete backward differential equations, which iteratively update the scalar value, gradient, and Hessian of V_k^i with respect to the belief. The introduced terminal constraints initialize these quantities from the terminal cost function.

Quadratic Approximation:

$$V_k^i(\bar{\mathbf{b}}_k + \delta \mathbf{b}_k) \approx V_k^i + V_{\mathbf{b},k}^{i,\top} \delta \mathbf{b}_k + \frac{1}{2} \delta \mathbf{b}_k^\top V_{\mathbf{bb},k+1}^i \delta \mathbf{b}_k$$

”**Corollary 2:** The discrete backward differential equations of the value functions V^i are:

$$V_k^i = Q^i + (Q_{\mathbf{u}}^i)^\top J_k + \frac{1}{2} J_k^\top Q_{\mathbf{uu}}^i J_k$$

$$V_{\mathbf{b},k}^i = (Q_{\mathbf{b}}^i)^\top + K_k^\top Q_{\mathbf{uu}}^i J_k + K_k^\top Q_{\mathbf{u}}^i + (Q_{\mathbf{ub}}^i)^\top J_k$$

$$V_{\mathbf{bb},k}^i = Q_{\mathbf{bb}}^i + K_k^\top Q_{\mathbf{uu}}^i K_k + K_k^\top Q_{\mathbf{ub}}^i + (Q_{\mathbf{ub}}^i)^\top K_k$$

with terminal constraints:”

$$V_l^i = c_l^i(\bar{\mathbf{b}}_l), \quad V_{\mathbf{b},l}^i = \left. \frac{\partial c_l^i(\mathbf{b})}{\partial \mathbf{b}} \right|_{\mathbf{b}=\bar{\mathbf{b}}_l},$$

$$V_{\mathbf{bb},l}^i = \left. \frac{\partial^2 c_l^i(\mathbf{b})}{\partial \mathbf{b}^2} \right|_{\mathbf{b}=\bar{\mathbf{b}}_l}$$

where c are the cost functions (according to notation, of agent i or concatenated of all agents), and $J \triangleq \hat{Q}_{uu} \hat{Q}_u$ is the feedforward term and $K_k \triangleq \hat{Q}_{uu} \hat{Q}_{bu}$ is the feedback term (see **Algorithm1** explanation).

Action-Value Function In practice, the action-value function Q_k^i is approximated quadratically by expanding both the belief dynamics and the cost-to-go around the nominal trajectory, as stated in *Theorem 2*. This yields the following eqns, where Q_k^i captures the scalar stage cost, the next-step value function, and curvature terms from the belief covariance. The vector $Q_{s,k}^i$ represents the gradient of Q_k^i with respect to the belief state, incorporating both deterministic and stochastic contributions. Finally, the Hessian $Q_{ss,k}^i$ encodes the local curvature of the action-value function in belief space.

Quadratic Approximation:

”**Theorem 2:** By linear expansion of the belief dynamics and quadratic expansion of the cost and value function, $Q_k^i(\mathbf{s}_k)$ is a quadratic of the form

$$Q_k^i(\bar{\mathbf{s}}_k + \delta \mathbf{s}_k) \approx Q_k^i + (Q_{s,k}^i)^\top \delta \mathbf{s}_k + \frac{1}{2} \delta \mathbf{s}_k^\top Q_{ss,k}^i \delta \mathbf{s}_k,$$

where:”

$$Q_k^i = c_k^i + V_{k+1}^i + \frac{1}{2} \sum_{j=1}^{n_x} (W_k^{(j)})^\top V_{bb,k+1}^i W_k^{(j)},$$

$$Q_{s,k}^i = c_{s,k}^i + g_{s,k}^\top V_{b,k+1}^i + \sum_{j=1}^{n_x} (W_{s,k}^{(j)})^\top V_{bb,k+1}^i W_k^{(j)},$$

$$Q_{ss,k}^i = c_{ss,k}^i + g_{s,k}^\top V_{bb,k+1}^i g_{s,k} +$$

$$\sum_{j=1}^{n_x} (W_{s,k}^{(j)})^\top V_{bb,k+1}^i W_{s,k}^{(j)}.$$

This approx serve the **backward pass**, which is central to the iLQG procedure, enabling optimal control updates by propagating cost information from the final time step to the start.

Bellman Eqn We know from class that for finite horizon cases, like discussed in paper, there are such optimality guarantees for choosing dynamic programming approach with the Bellman eqn recursive setup. Such method theoretically consists of solving these set of eqns iteratively:

”best Response” POMDP methodology, which requires a

recursive setup:

$$\begin{aligned} &\rightarrow V_l^i(\mathbf{b}_l) = c_l^i(\mathbf{b}_l) \\ &\rightarrow Q_k^i(\mathbf{b}_k, \mathbf{u}_k) = c_k^i(\mathbf{b}_k, \mathbf{u}_k) + \\ &+ \mathbb{E}_{\mathbf{z}_{k+1}} [V_{k+1}^i(\psi(\mathbf{b}_k, \mathbf{u}_k, \mathbf{z}_{k+1}))] \\ &\rightarrow V_k^i(\mathbf{b}_k) = \min_{\mathbf{u}_k^i} Q_k^i(\mathbf{b}_k, \mathbf{u}_k) \\ &\rightarrow \pi_k^i(\mathbf{b}_k) = \arg \min_{\mathbf{u}_k^i} Q_k^i(\mathbf{b}_k, \mathbf{u}_k) \end{aligned}$$

And in paper **Algorithm1** was introduced as a solver which uses the principles of such Bellman eqn.

3.4 Formal Problem Statement- Nash Equilibrium Convergence

The integration of the Nash equilibrium within the presented framework is formalized in **Theorem 1**, which states that a necessary condition for a local Nash equilibrium is the vanishing gradient of the action-value function with respect to each agent’s control,

$$\frac{\partial Q_k^i(\mathbf{b}_k, \mathbf{u}_k)}{\partial \mathbf{u}_k^i} = 0,$$

for all agents $i \in \{1, 2, \dots, N\}$ and time steps $k \in \{0, 1, \dots, l-1\}$. This condition ensures that no unilateral deviation by any agent can reduce its cost-to-go. In practice, this is embedded within an iterative loop, where control policies are updated until the Bellman optimality criterion

$$\left| Q^i(\mathbf{b}_{\text{new}}, \mathbf{u}_{\text{new}}) - Q^i(\bar{\mathbf{b}}, \bar{\mathbf{u}}) \right| < \epsilon$$

is satisfied, guaranteeing convergence toward a local Nash equilibrium, and also guaranteed as mentioned in finite horizon cases.

Quadratic static problem In paper, it was mentioned and developed why solving the Bellman recursion simultaneously for all agents defines a static game, a game at each stage k of the backward-pass. Thus, at time stage k, each agent i solves the quadratic problem:

$$\delta \mathbf{u}_k^{i,*} = \arg \min_{\delta \mathbf{u}_k^i} \mathbf{Q}_{s,k}^{i,\top} \delta \mathbf{s}_k + \frac{1}{2} \delta \mathbf{s}_k^\top \mathbf{Q}_{ss,k}^i \delta \mathbf{s}_k$$

where the solution to such quadratic game, for the control perturbations around the nominal converged Nash controls also proposed as such:

$$\delta \mathbf{u}_k^* = -\hat{\mathbf{Q}}_{uu}^{-1} \left(\overbrace{\hat{\mathbf{Q}}_u}^{\text{feed forward}} + \overbrace{\hat{\mathbf{Q}}_{ub} \delta \mathbf{b}_k}^{\text{feedback}} \right)$$

where $\hat{\mathbf{Q}}_{uu}, \hat{\mathbf{Q}}_{ub}, \hat{\mathbf{Q}}_u$ are defined as

$$\hat{\mathbf{Q}}_{uu} = \begin{bmatrix} Q_{u^1 u^1}^1 \\ Q_{u^2 u^2}^2 \\ \vdots \\ Q_{u^N u^N}^N \end{bmatrix}, \quad \hat{\mathbf{Q}}_{ub} = \begin{bmatrix} Q_{u^1 b}^1 \\ Q_{u^2 b}^2 \\ \vdots \\ Q_{u^N b}^N \end{bmatrix}, \quad \hat{\mathbf{Q}}_u = \begin{bmatrix} Q_{u^1}^1 \\ Q_{u^2}^2 \\ \vdots \\ Q_{u^N}^N \end{bmatrix}$$

$$\mathbf{Q}_s^i = \begin{bmatrix} \mathbf{Q}_b^i \\ \mathbf{Q}_{u^1}^i \\ \vdots \\ \mathbf{Q}_{u^N}^i \end{bmatrix}, \quad \mathbf{Q}_{ss}^i = \begin{bmatrix} \mathbf{Q}_{bb}^i & \mathbf{Q}_{bu^1}^i & \cdots & \mathbf{Q}_{bu^N}^i \\ \mathbf{Q}_{u^1 b}^i & \mathbf{Q}_{u^1 u^1}^i & \cdots & \mathbf{Q}_{u^1 u^N}^i \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{Q}_{u^N b}^i & \mathbf{Q}_{u^N u^1}^i & \cdots & \mathbf{Q}_{u^N u^N}^i \end{bmatrix}.$$

EKF state estimator Extending to **EKF** when re-linearizing nonlinear dynamics, selecting appropriate nominal trajectory, and estimating the states (recall : same eqns as in Classic EKF, when Perturbations are involved in Background), the state estimation of each agent will be done as such:

$$\begin{aligned} \text{A-Posteriori} \quad \hat{\mathbf{x}}_{k+1} &= \overbrace{f(\hat{\mathbf{x}}_k, \mathbf{u}_k, \mathbf{0})}^{\text{Prediction}} + \overbrace{\mathbf{K}_k}^{\text{Optimal Gain}} \overbrace{(\mathbf{z}_{k+1} - h(f(\hat{\mathbf{x}}_k, \mathbf{u}_k, \mathbf{0}), \mathbf{0}))}^{\text{Innovation}} \\ \text{A-Posteriori} \quad \hat{\Sigma}_{k+1} &= \overbrace{\hat{\Gamma}_{k+1}}^{\text{Prediction}} - \overbrace{\mathbf{K}_k}^{\text{Optimal Gain}} \cdot \overbrace{\mathbf{H}_k \hat{\Gamma}_k}^{\text{Innovation}} \end{aligned}$$

with corresponding matrices defined by

$$\mathbf{\Gamma}_{k+1} = \mathbf{A}_k \mathbf{\Sigma}_k \mathbf{A}_k^\top + \mathbf{M}_k \cdot \mathbf{I} \cdot \mathbf{M}_k^\top$$

optimal Kalman Gain :

$$\mathbf{K}_k = \mathbf{\Gamma}_{k+1} \mathbf{H}_k^\top \left(\mathbf{H}_k \mathbf{\Gamma}_{k+1} \mathbf{H}_k^\top + \mathbf{N}_k \cdot \mathbf{I} \cdot \mathbf{N}_k^\top \right)^{-1}$$

Linearization Jacobians :

$$\mathbf{A}_k = \frac{\partial f}{\partial \mathbf{x}}(\hat{\mathbf{x}}_k, \mathbf{u}_k, \mathbf{0}), \quad \mathbf{M}_k = \frac{\partial f}{\partial \mathbf{m}}(\hat{\mathbf{x}}_k, \mathbf{u}_k, \mathbf{0})$$

$$\mathbf{H}_k = \frac{\partial h}{\partial \mathbf{x}}(f(\hat{\mathbf{x}}_k, \mathbf{u}_k, \mathbf{0}), \mathbf{0}), \quad \mathbf{N}_k = \frac{\partial h}{\partial \mathbf{n}}(f(\hat{\mathbf{x}}_k, \mathbf{u}_k, \mathbf{0}), \mathbf{0})$$

$$\left(\mathbf{m}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \mathbf{n}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \right)$$

3.5 Levenberg–Marquardt Regularizations

In Newton Like problems, to ensure convergence regardless of initial conditions, such regularizations were introduced: which affected directly the second orders of the long term influence of certain controls in the process of convergence to Nash controls:

1. **Control Regularization:** The control regularization is achieved by adding a diagonal term of magnitude μ_u to the diagonal of $\hat{Q}_{uu}$, yielding :

$$\tilde{Q}_{uu}^i = \hat{Q}_{uu}^i + \mu_u \mathbf{I}.$$

2. **Belief Regularization:** To ensure that the updated belief trajectory does not deviate too far from the previous iteration, we introduce a scheme that penalizes deviations from beliefs rather than controls with parameter μ_b :

$$\begin{aligned} \hat{Q}_{ss,k}^i &= c_{ss,k}^i + g_{s,k}^\top \left(V_{bb,k+1}^i + \right. \\ &\quad \left. + \mu_b I g_{s,k} + \sum_{i=1}^n W_{s,k}^{(j),\top} \left(V_{bb,k+1}^i + \mu_b I \right) W_{s,k}^{(j)} \right). \end{aligned}$$

***Potential Differential Games** This theory is not governed in the main paper discussed [1], but in the related work paper [14].

In general, by definition, Differential Game tuple considered as:

$$\Gamma_{\mathbf{x}_0}^T \triangleq (N, \{U_i\}_{i=1}^N, \{J_i\}_{i=1}^N, f),$$

What it really gives us to move from Differential Game (DG) to Potential Differential Game (PDG)? Leveraging the special **inherent** structure of multi-agent interactions to generate interactive trajectories **by simply solving a single optimal control problem** - the OCP associated with:

minimizing the **potential function** of the differential game.

When can we move from DG to PDG? Requires mutual coupling costs which are **symmetric**, and **Decoupled Dynamics**. DG is PDG (according to *Definition 2* in [14]) "if there exists an OCP whose solutions are **Nash Equilibria (NE)** of $\Gamma_{\mathbf{x}_0}^T$ ". The

generic cost structure, associated with PDG:

$$\begin{aligned} J_i(\mathbf{x}_0, u_1, \dots, u_N) &= \int_0^T \underbrace{p(\mathbf{x}(t), \mathbf{u}(t), t) + c_i(\mathbf{x}_{-i}(t), \mathbf{u}_{-i}(t), t)}_{L_i(\mathbf{x}(t), \mathbf{u}(t), t), \text{Running costs}} dt \\ &\quad + \underbrace{\bar{s}(\mathbf{x}(T)) + s_i(\mathbf{x}_{-i}(T))}_{S_i(\mathbf{x}(T)), \text{terminal costs}} \end{aligned}$$

In [14], it was **proven** that open loop control that minimizes the following OCP is **NE**:

$$\min_{\mathbf{u}(\cdot)} \int_0^T \underbrace{p(\mathbf{x}(t), \mathbf{u}(t), t) dt + \bar{s}(\mathbf{x}(T))}_{\text{potentials}} \quad ; \quad \text{s.t.} \quad \underbrace{\dot{x}_i(t) = f_i(x_i(t), u_i(t), t)}_{\text{Decoupled Dynamics}}$$

Potential reduction $\forall$ agents (simply solve **single OCP**).

*instead of **N** OCP's according to the paper discussed.

Understanding the **Potentials** structure in our case studies to follow, this is the potential general structure:

$$\begin{aligned} p(\mathbf{x}, \mathbf{u}, t) &= \sum_{i=1}^N C_i^{tr}(x_i, u_i) + \sum_{1 \leq i < j} C_{ij}^a(x_i, x_j), \\ \bar{s} &= \sum_i C_{i,T}^{tr}(x_i), \end{aligned} \tag{4}$$

How much (qualitatively) **a bunch** of agents are **on track** with their nominal trajectory, while avoiding **mutual** collisions?

3.6 Assumptions and Scope

Before presenting **Algorithm1** as the discussed solver in paper, we found it important to emphasize the main assumptions made on it:

- A1 Gaussian beliefs:** beliefs remain Gaussian after EKF updates.
- A2 Smooth dynamics:** f and h are differentiable; linearisation errors stay small.
- A3 First-order beliefs:** planning and prediction are limited to first-order beliefs and not deeper (agents share beliefs about each other).
- A4 Common knowledge:** all agents know same cost, dynamics and observation models.

We discuss how relaxing these assumptions could enable decentralised or non-Gaussian variants in Section 5.

3.7 Algorithm 1 – Belief–Space iLQG

Algorithm 1 Belief–Space iLQG for Multi-Agent Stochastic Dynamic Games

Require: Initial belief b_0 (EKF estimate), nominal trajectories $\{\bar{b}_k\}_{0:T}$, $\{\bar{u}_k\}_{0:T-1}$, tolerances $\varepsilon_J, \varepsilon_Q$, initial LM regs

$$\mu_u = \mu_b = 10^{-3}$$

- 1: $J_{\text{prev}} \leftarrow +\infty$ (total cost from previous iteration); $\Delta_Q \leftarrow +\infty$ ($\Delta_Q \triangleq |Q^i(\mathbf{b}_{\text{new}}, \mathbf{u}_{\text{new}}) - Q^i(\bar{\mathbf{b}}, \bar{\mathbf{u}})|$)
 - 2: **repeat** $\triangleright$ Forward pass: EKF rollout and local models
 - 3: **for** $k = 0$ **to** $T - 1$ **do**
 - 4: Propagate each agent’s belief (μ_k, Σ_k) with EKF to get b_{k+1} ; cache Jacobians A_k, H_k and noise maps M_k, N_k
 - 5: Form linearized belief dynamics $b_{k+1} \approx g_k + G_{b,k} \delta b_k + G_{u,k} \delta u_k$
 - 6: Quadraticize costs for all agents: $Q_k^i(\delta b_k, \delta u_k) = c_k^i + q_{s,k}^{i\top} \delta s_k + \frac{1}{2} \delta s_k^\top Q_{ss,k}^i \delta s_k$, $\delta s_k = \begin{bmatrix} \delta b_k \\ \delta u_k \end{bmatrix}$
 - 7: **end for** $\triangleright$ Backward pass: solve quadratic game at each k with LM regularization
 - 8: Initialize terminal value approximations for each agent i : $V_T^i, V_{b,T}^i, V_{bb,T}^i$ from $\ell_T^i(b_T)$
 - 9: **for** $k = T - 1$ **down to** 0 **do**
 - 10: Build block terms $Q_{bb,k}^i, Q_{bu,k}^i, Q_{uu,k}^i, q_{b,k}^i, q_{u,k}^i$ using $(G_{b,k}, G_{u,k})$ and $(V_{b,k+1}^i, V_{bb,k+1}^i)$
 - 11: **Apply LM regularization:**
 - 12: $\tilde{Q}_{uu,k}^i \leftarrow Q_{uu,k}^i + \mu_u I$; $\tilde{Q}_{ss,k}^i \leftarrow Q_{ss,k}^i$ but with $V_{bb,k+1}^i \mapsto V_{bb,k+1}^i + \mu_b I$
 - 13: Solve the static quadratic game for joint affine policy
$$\pi_k : \delta u_k = j_k + K_k \delta b_k$$
 - 14: where K_k and j_k are the unique Nash solution of the block KKT system built from all $\{\tilde{Q}_{\cdot,k}^i\}_{i=1}^N$
 - 15: Update value function coefficients for each agent:
$$\begin{aligned} V_k^i &\leftarrow Q_k^i + (Q_{u,k}^i)^\top j_k + \frac{1}{2} j_k^\top Q_{uu,k}^i j_k \\ V_{b,k}^i &\leftarrow Q_{b,k}^i + (Q_{bu,k}^i)^\top j_k + K_k^\top Q_{u,k}^i + K_k^\top Q_{uu,k}^i j_k \\ V_{bb,k}^i &\leftarrow Q_{bb,k}^i + (Q_{bu,k}^i)^\top K_k + K_k^\top Q_{ub,k}^i + K_k^\top Q_{uu,k}^i K_k \end{aligned}$$
 - 16: **end for** $\triangleright$ Forward rollout under updated feedback policies
 - 17: For $k = 0 \dots T - 1$: set $u_k \leftarrow \bar{u}_k + j_k + K_k (b_k - \bar{b}_k)$; roll out EKF belief dynamics to get $\{b_k\}_{0:T}$
 - 18: Compute per-agent action-values Q_{new}^i along rollout; set $\Delta_Q \leftarrow \max_i |Q_{\text{new}}^i - Q_{\text{old}}^i|$
 - 19: **if** $\Delta_Q < \varepsilon_Q$ **and** total cost J decreases **then**
 - 20: **Accept:** $\{\bar{b}_k, \bar{u}_k\} \leftarrow \{b_k, u_k\}$; $\mu_u \leftarrow 0.5 \mu_u$, $\mu_b \leftarrow 0.5 \mu_b$
 - 21: **else**
 - 22: **Reject:** $\mu_u \leftarrow 10 \mu_u$, $\mu_b \leftarrow 10 \mu_b$; **continue** (retry with stronger regularization)
 - 23: **Convergence check:** stop if $|J - J_{\text{prev}}| < \varepsilon_J$ **and** $\Delta_Q < \varepsilon_Q$; else $J_{\text{prev}} \leftarrow J$
 - 24: **converged**
 - 25: **return** Robot policy $\{\pi_k\}_{0:T-1}$, predicted opponent policies $\{\pi_k^{-1}\}$, and trajectories $\{\bar{b}_k\}, \{\bar{u}_k\}$
-

4 Main contribution

This section unpacks the core contribution of Schwarting *et al.*—an iterative Linear–Quadratic–Gaussian (iLQG) solver lifted to *Gaussian belief space* and embedded in a stochastic dynamic game. We first reconnect to the problem setup, then present the algorithm in full, discuss runtime, provide intuition and a parallel simulation, and finally outline limitations.

Algorithm overview: Algorithm 1 iterates five macro-steps: (1) *initialise* a nominal belief–control trajectory $(\bar{b}_0, \bar{u}_{0:T-1})$; (2) *forward EKF pass* to collect Jacobians; (3) *backward Riccati pass* that solves a quadratic dynamic game and yields feed-forward j_k and feedback K_k ; (4) *forward rollout* under the new policy; (5) *regularisation check* – reduce the LM terms if the cost drops, or increase them and retry otherwise. Repeat until $|Q^i(\mathbf{b}_{\text{new}}, \mathbf{u}_{\text{new}}) - Q^i(\bar{\mathbf{b}}, \bar{\mathbf{u}})| < \epsilon$ (Nash equilibrium convergence).

4.1 Runtime and Scalability Analysis

Agents N	State dim. n	Horizon T	Time per iter. $\sim$ [ms]
2	4	50	18
5	6	50	94
10	6	50	580

Table 1: Empirical runtime per iLQG iteration

Parallelisation and Memory Scaling. The computation of the Jacobians for each agent is inherently parallelisable, since these evaluations are independent across agents. Leveraging GPU batching, the effective runtime complexity can be reduced from the nominal $\mathcal{O}(TN^2)$ to approximately $\mathcal{O}(TN)$, where T denotes the planning horizon and N the number of agents. This is particularly beneficial in the belief–space iLQG framework, in which each iteration requires repeated linearisation of the belief dynamics and cost functions for all agents. From a memory perspective, the dominant requirement arises from storing the value–function Hessians $V_{bb,k}$ and the feedback gains K_k for all $k \in \{0, \dots, T-1\}$ and for all agents, which in a naïve implementation scales as $\mathcal{O}(TN^2)$. By exploiting low–rank Kronecker structure inherent in certain belief–space dynamics, this storage complexity can be reduced to $\mathcal{O}(TN)$, thereby enabling the algorithm to address larger multi–agent problems without exceeding practical memory limits.

4.2 Active Surveillance

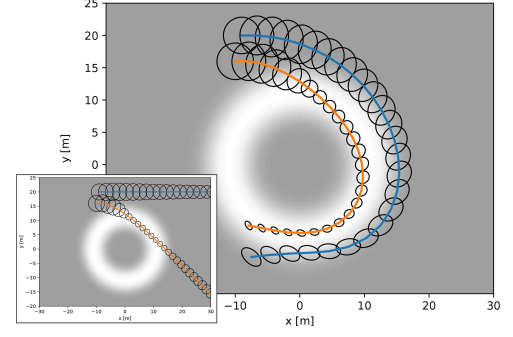


Figure 1: The seeker (blue) minimises the evader’s positional uncertainty while keeping chance constraints below 1%.

Setup. 2-agent planar pursuit–evasion, dynamics as in [1].

Result. BSP shrinks the evader’s 2σ ellipse by **>50 %** versus a certainty–equivalent planner.

Has Decoupled Dynamics? **Yes.**

4.3 Guide-Dog

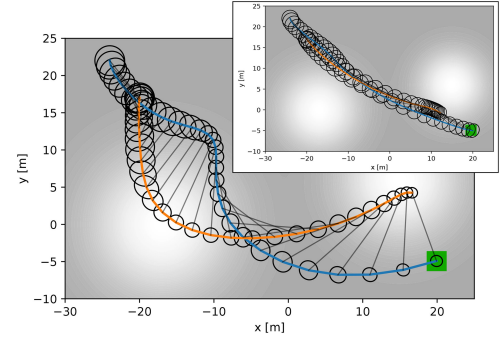


Figure 2: Blind agent (green) is led safely while information gain reduces final position error.

Setup. Leader–follower with a virtual leash, EKF range–bearing observations.

Result. BSP cuts the blind agent’s terminal error by **40 %**.

Has Decoupled Dynamics? **No.**

4.4 Autonomous Racing

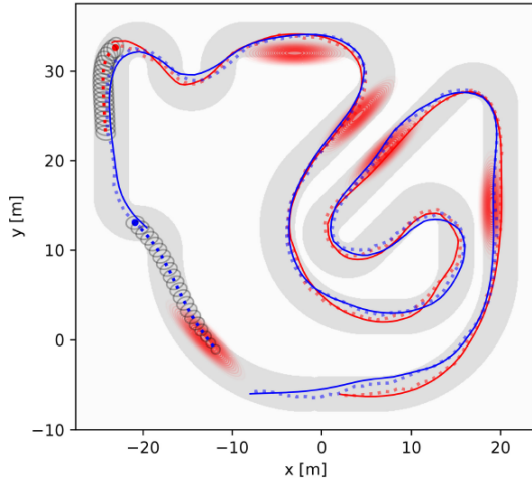


Figure 3: The blue agent starts with a disadvantage but is equipped with better acceleration and capable of moving faster through corners than the red agent. Solid lines: optimised trajectories; ellipses: 2σ positional belief. Collision probability remains below 1% throughout.

The figure illustrates how belief-iLQG produces an anticipatory swerve by Agent 2, creating space for Agent 1’s overtake while controlling estimation uncertainty.

Result. BSP + dynamic game wins **44 %** more races and incurs $10\times$ fewer collisions than the non-BSP variant.

Has Decoupled Dynamics? Yes.

4.5 Quantitative Results & Baseline Comparison

Scenario	Metric	BSP-DG	Baseline
Active Surveillance	σ_{evader} shrink	50 %	12 %
Guide-Dog	Final pos. error	0.38 m	0.64 m
Racing	Win ratio	+44 %	–
Racing	Collisions / 100 m	0.01	0.10

Table 2: Headline numbers

Table 2 summaries the key quantitative outcomes across the three evaluated scenarios, comparing the proposed BSP–DG framework against the baseline. In the *Active Surveillance* task, the

method achieves a substantial reduction in the evader’s belief covariance, with σ_{evader} shrinking by 50% compared to only 12% for the baseline, indicating significantly improved localisation and tracking efficiency. In the *Guide–Dog* scenario, the final positional error is reduced to 0.38 m, outperforming the baseline error of 0.64 m by a wide margin. In the *Racing* scenario, BSP–DG achieves a 44% higher win ratio than the opponent and reduces collision frequency from 0.10 to 0.01 collisions per 100 m, demonstrating both superior competitiveness and enhanced safety. These improvements are consistent across tasks, underscoring the robustness and adaptability of the proposed belief–space dynamic game formulation.

4.6 Qualitative Insights & Baselines Comparison

Anticipatory Behaviour. The feedback gain K_k inherently encodes each agent’s anticipated response to the actions of others, resulting in trajectories that reflect implicit *negotiation* between agents. This leads to smoother, more cooperative motion patterns compared to open–loop MPC, where such mutual adaptation is absent.

Noise–Aware Control. Uncertainty enters the optimisation explicitly through the covariance–weighted cost term L_{bb} and, consequently, through the feedback law K_k . Elevated uncertainty increases the predicted cost–to–go, encouraging more conservative control actions that mitigate risk. In contrast, classical differential games with purely deterministic dynamics neglect these effects, leading to behaviour that is less robust to stochastic disturbances.

Method	Uncertainty	Strategic	Complexity
Standart LQG	Yes	No	$O(T)$
MPC (replan)	Partial	Partial	$O(TN)$
Dec-POMDP exact	Yes	Yes	$\exp(T, N)$
Belief-iLQG	Yes	Yes	$(O(TN^3))$ (empirical) $O(\ell N^7 n_x^6)$ (worst-case))

Table 3: Qualitative comparison with common baselines.

4.7 Summary and Transition

We provided a detailed mathematical review of the belief-space iLQG solver, laid out its algorithmic steps, analysed runtime, and

offered qualitative insight through comparison and simulation results made in paper. The solver, suggested in paper, strikes a rare balance: *robust stochastic optimisation with strategic awareness at polynomial cost*. Section 5 will critically assess these strengths in light of modelling assumptions and real-world deployment constraints.

5 Discussion and Critical Thinking

5.1 Strengths of the Belief-Space iLQG Framework

The proposed framework advances the state of the art in three key dimensions. *First*, it unifies uncertainty-aware decision making with strategic multi-agent reasoning: the feedback gains K_k reflect both belief covariances and the anticipated reactions of other agents, a capability absent in classical LQG and deterministic game solvers. *Second*, despite a theoretical *worst-case* complexity of $O(\ell N^7 n_x^6)$, exploiting the block-diagonal structure of $Q_{uu,k}$ reduces empirical scaling to $O(TN^3)$ (Sec. 4.1), enabling real-time performance for $N \leq 5$. *Third*, the modular architecture supports any EKF variant (e.g., invariant EKF on SE(3)), broadening the applicability across sensing modalities and platform types.

5.2 Limitations and Fragile Assumptions

Gaussianity. All beliefs are constrained to remain Gaussian; multi-modal posteriors from bearings-only measurements or map symmetries violate this assumption and can degrade performance.

Local optimality. As with standard iLQG, convergence is to a *local* Nash equilibrium; poor nominal trajectories may lead to suboptimal solutions. The Levenberg–Marquardt regularisation alleviates but does not eliminate this sensitivity.

Common knowledge model. Assumption A4. While the algorithm can, in principle, adapt to the other agents’ responses through the coupled backward pass, it assumes that the structure of their dynamics and cost functions **is known and fixed**. In partially competitive or adversarial domains, even small discrepancies in these models can lead to qualitatively incorrect equilibrium behaviours. Robust opponent-model adaptation or online learning of cost weights would be required for safety-critical use.

5.3 Sensitivity Analysis and Robustness

Considering the Regularization tuning needed, The Levenberg–Marquardt regularisation is vital for ensuring positive definiteness of $Q_{uu,k}$, yet it introduces an additional hyper-parameter scheduling problem. Over-regularisation degrades performance by damping feedback K_k excessively; under-regularisation risks numerical instability. Adaptive rules that respond to curvature information in $Q_{uu,k}$ could replace hand-tuned schedules.

5.4 Scalability and Computational Trade-offs

Runtime scales cubically with the number of agents since each $Q_{uu,k}$ inversion involves an $m \times m$ block per agent. For $N > 10$, two mitigations are effective: (i) executing the backward pass to approach $O(TN^2)$, and (ii) approximating $Q_{uu,k}^{-1}$ with a block-diagonal structure plus rank- r low-rank corrections, yielding $O(N^2 r)$ per step. Memory use **grows as $O(TN^2)$, but low-rank storage of $Q_{uu,k}$ can reduce the constant factor**.

5.5 Ethical and Safety Considerations

- **Strategic exploitation.** Malicious agents could misreport or manipulate covariance estimates to induce unsafe responses; countermeasures include trust metrics or conservative covariance inflation.
- **Overconfidence.** Risk under unmodelled uncertainty, where The Gaussian belief assumption not only underrepresents tail risks but also limits the planner’s ability to handle discrete or multimodal uncertainty (e.g., agent identity ambiguity, mode switches). In scenarios with high-impact rare events, risk-sensitive extensions or chance-constrained formulations would better capture safety margins.
- **Transparency.** Human interpretability and trust - While the feedback gains K_k are analytically derived, their connection to intuitive “rules of the road” or cooperative/competitive strategies is opaque to non-experts. In mixed human–autonomy settings, lack of interpretability may limit trust and hinder adoption. Post-hoc policy summarisation or explainable-policy surrogates could help bridge this gap.

5.6 *Additional Critical Considerations

Sensitivity to belief linearisation. The EKF linearisation of the belief dynamics (A_k, H_k) is carried out around the nominal trajectory, and errors in this linearisation directly propagate into both the forward simulation and the quadratic game matrices. Highly non-linear dynamics, discontinuous observations, or sudden topology changes in the environment (e.g., occlusions) can lead to linearisation error that undermines convergence. Non-linear filtering variants (UKF, particle filters) could mitigate this at additional cost.

Computational coupling in multi-agent settings. Although the complexity is polynomial, the backward pass in the coupled Riccati recursion scales poorly with the number of agents due to dense $Q_{uu,k}$ blocks. For larger teams, this creates a hard trade-off between solution fidelity (full coupling) and scalability (approximate decoupling). Designing structured approximations that preserve critical cross-agent couplings while reducing block size remains an open challenge. [In the next chapter, we'll discuss a solution approach we tried to phrase by ourselves, which reduces significantly the number of optimization problems to solve via potential collective costs.](#)

Interaction with hard constraints. The framework operates on smooth cost functions and soft penalties; in high-stakes domains with strict safety boundaries (e.g., collision constraints, no-fly zones), it relies on penalty shaping to approximate hard constraints. Without explicit feasibility handling (e.g., via augmented Lagrangian or projection), there is no guarantee of strict constraint satisfaction, particularly under estimation errors.

5.7 Future Research Directions

- (a) **Non-Gaussian filtering.** Replace EKF with unscented or particle iLQG [5] to handle multimodal beliefs.
- (b) **Learned Jacobians.** Employ online-trained neural surrogates for A_k, B_k to adapt to unknown or time-varying dynamics.
- (c) **Distributed optimisation.** Dual-decomposition across agents to relax Assumption A3 and enable scaling to larger swarms with gradient exchange only.
- (d) **Higher-order interactive beliefs.** Extend beyond first-order reasoning to second-order ("I believe that you believe ...") or level- k hierarchies, with truncated or particle-based approximations to manage complexity.

6 Own Extension

In this part, we would like to present our own extension and practical demonstration to the themes presented in the Main paper [1] and in the Relative work [14]. As a direct continue to the case study presented in [1], *Autonomous Racing*, we built a complete simulation of two physical racing cars, modeled as points with mass, which compete in a noncooperative way to finish the lap 1st. *We'll mention that the simulations made were eased version of the proposals in paper, but demonstrate later own our own main takeaway in considering between the two approaches. With that in mind, the solutions we converged to are not the optimal according to the theory discussed. -We provide the code in the next Github link : [Click here to reach the Code](#)

6.1 The modeling

The Track is circular, with *light-zones* along it, as opportunities for the agents to reduce the uncertainty about their own states, but also harm them when they let the other agent know about their states. Assuming physical *CarModels*, which specify each car drag and friction coefficients with the track. For each agent on track:

The states considered: cartesian locations and axial velocities:

$$\mathbf{x}^{(i)} = [x^{(i)}, y^{(i)}, v_x^{(i)}, v_y^{(i)}]$$

The controls: not steering as mentioned in paper, but consequential physical tangential and radial accelerations:

$$\mathbf{u}^{(i)} = [a_{t,k}^{(i)}, a_{r,k}^{(i)}]$$

The continuous dynamics:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} v_x \\ v_y \\ a_t(t) \cdot \hat{t}_x(\mathbf{x}) + a_r(t) \cdot \hat{r}_x(\mathbf{x}) \\ a_t(t) \cdot \hat{t}_y(\mathbf{x}) + a_r(t) \cdot \hat{r}_y(\mathbf{x}) \end{bmatrix}$$

Running costs, for each agent:

$$\begin{aligned} \ell_i(\mathbf{x}_i, \mathbf{u}_i, \mathbf{x}_j, \Sigma_i, \Sigma_j) &= w_v \cdot (v_i - v_{\text{des},i})^2 \\ &+ w_r \cdot (r_i - R_{\text{inner}})^2 \\ &+ w_c \cdot \exp\left(-\frac{1}{2} \cdot \frac{\Delta \boldsymbol{\mu}_{ij}^\top \Sigma_{ij}^{-1} \Delta \boldsymbol{\mu}_{ij}}{\lambda_{\text{collision}}}\right) \end{aligned}$$

Terminal costs:

$$\phi_i(\mathbf{x}_i) = w_T \cdot (s_{\text{goal}} - s_i)^2$$

Process and Observation noises:

$$m \sim \mathcal{N}(0, Q) \quad ; \quad n \sim \mathcal{N}(0, R)$$

6.2 Iterative best-Response simulation

After setting the model right, we've constructed the code to implement the idea behind **Algorithm1**, suggested in [1]. Such simulation result:

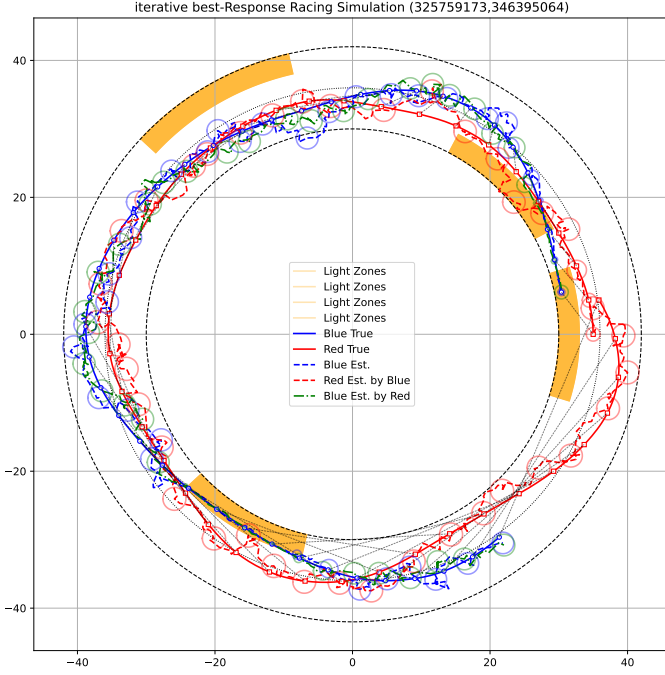


Figure 4: The blue agent, at first, has the certainty (spawns at light) and placement advantages, but equipped with reduced performance capabilities, such as higher drag and friction. The red at first observes the light but doesn't manage to reach it (probably because it overshoots the tangential acceleration control)

where were drawn: light zones- in yellow, bold lines- indicating the true physical trajectories of each agent, broken lines- indicating the belief trajectory of each agent on its opponent (red and blue are zero order beliefs, and blue is first order belief of red on blue), covariance ellipses- every 10 time steps, broken grey lines- indicating the relative location between the agents.

6.2.1 Analysis

We would like to present additional analysis we did on the simulation which strengthens and emphasizes the expected **Cognitive Reasoning** of the agents.

I First analysis we did was looking for some correlation between the acceleration control of the agents and the compared distance

of each one from the centerline (the nominal center, which is considered as the neutral trajectory to follow).

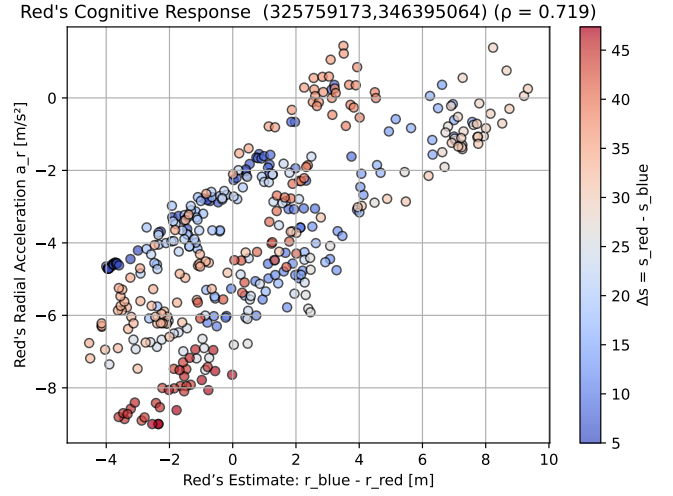


Figure 5: How does Red adapt its turning behavior (radial acceleration) depending on where it believes Blue is located relative to itself?

For example: Red's estimate- Blue is further inside the track, closer to the inner boundary ($(r_{blue} - \overbrace{r_{centerline}}^0) - (r_{red} - r_{centerline}) < 0$), which correspond in this case to the lower points in the graph. Corresponding Behavior: Red increases its inward turning (more negative a_r). **Assumed cognitive behavior:** If Blue is on the inside, Red may close up (turn inwards) to opening a gap while not risking collision — an Attacking strategy. Δs indicates progress along the track.

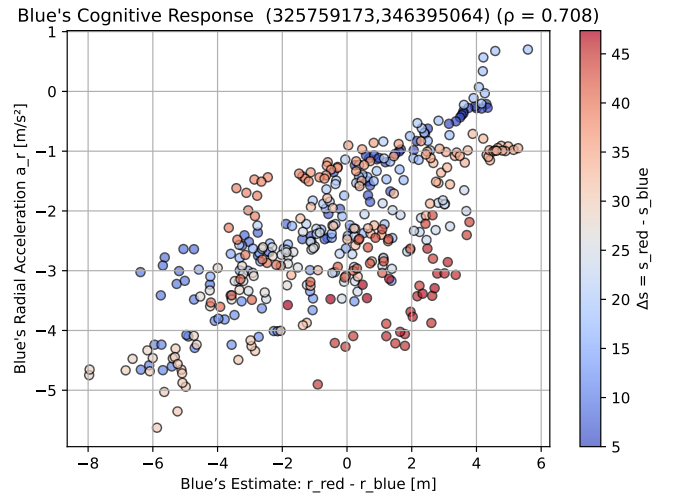


Figure 6: How does Blue adapt its turning behavior (radial acceleration) depending on where it believes Red is located relative to itself? Same cognitive behavior observed.

We ran multiple simulations in the configuration presented, and we observed **clearly meaningful values indicating there is a strong linear correlation between the two parameters discussed**, which means that the reasoning mentioned is . In the simulation presented, such values were in the range $\rho \approx 0.7$.

II Second analysis done in order to look for *Common Knowledge* assumption laying inside the deeper layers behind the reasoning of the agents. As mentioned, in our case, Common knowledge would imply that both agents have perfect, symmetric, and accurate knowledge of each other's state — especially radial position, since it's critical for avoiding collisions and making strategic decisions.

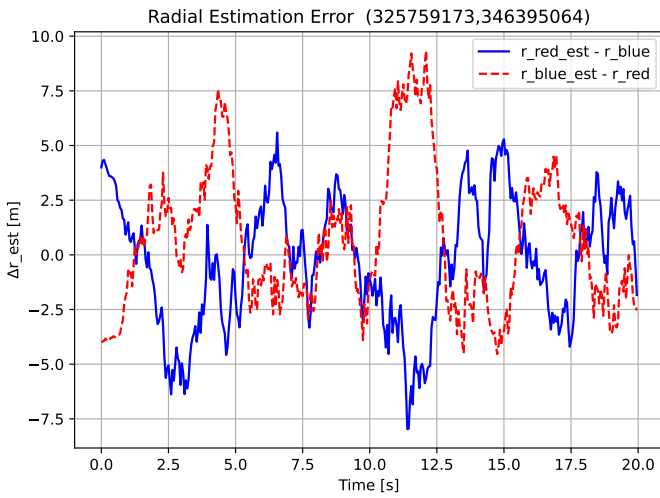


Figure 7: *Radial location Estimation Error over time for both agents — directly speaks to the validity (or breakdown) of the common knowledge assumption in the multi-agent racing scenario.*

We can clearly see that the graphs are **symmetric in average around the centerline**, while also assuming, and observing, that the estimations on each agent state are fluctuating quite similar around the ground truths, but also we'll account for cases where it is not symmetric. Such cases may rise due to the Asymmetry in the track itself, and the uncertainty differences it may rise. For example, around 2.6 seconds into the simulation, the uncertainty about the Red's states is quite big compared to the Blue's, which spawned in light and just departed from it, leaving him with high certainty on its states and the Red also on the Blue's state. Such scenario leads to fluctuation seen on the Red's Belief, and the shift downwards seen in graph. Many more cognitive analysis can be done, and it was important

for us to observe such proofs for decision making and reasoning according to the costs constructed.

6.3 Improvement - potential Differential Game simulation

In addition to the simulation made earlier, **another simulation was made**, in order to move from the solution approach mentioned before, where Nash-Equilibrium convergence done by letting each agent optimize its own control assuming the other agent's control is fixed — then iterating until strategies stabilize (iterative best-Response), to solution approach where Nash-Equilibrium convergence is done through a **cooperative optimization** of a **single** global potential function. The significant meaning of moving to such approach is that at each time instant in the simulation, *only single (iterative, optional) optimization problem needs to be solved, instead of N, as the number of agents involved, iterative optimization problems*. That was our motivation when we tried to show similar results achieved using that approach.

6.3.1 The modeling

Moving to such approach, We'll modify and reconstruct the cost function, **one collaborative cost function**, as such: Running Cost:

$$\begin{aligned} \ell_{\text{pot}} = & w_v(v_{\text{blue}} - v_{\text{des,blue}})^2 \\ & + w_v(v_{\text{red}} - v_{\text{des,red}})^2 \\ & + w_r(r_{\text{blue}} - R_{\text{inner}})^2 \\ & + w_r(r_{\text{red}} - R_{\text{inner}})^2 \\ & + w_c \exp\left(-\frac{1}{2} \cdot \frac{\Delta\mu^\top \Sigma^{-1} \Delta\mu}{\lambda_{\text{collision}}}\right) \end{aligned}$$

consisting of all the additive components of tracking and mutual influence between themselves.

Terminal Potential Cost:

$$\phi_{\text{pot}} = w_T(s_{\text{goal}} - s_{\text{blue}})^2 + w_T(s_{\text{goal}} - s_{\text{red}})^2$$

collectively accounting for best final placement of the agents.

6.3.2 Assumptions

Recalling the necessary conditions and assumptions, for implementing the theory on our case study, the two main conditions were:

- *Symmetric mutual coupling costs* - as we've already discussed earlier in the analysis chapter, where we can clearly see *Common Knowledge* assumption taking significant place in reasoning, which we can relate directly to the resultant **symmetric anti-collision** cost construction of all the agents, and the symmetric anti collisions with the wall.
- *Decoupled Dynamics* - each agent is **independant** of the other agents dynamics in the environment (consistently with the notation $\square_{(i)}$).

Both take place in our case.

6.3.3 Simulation & Analysis

Such parallel candidate simulation of that approach:

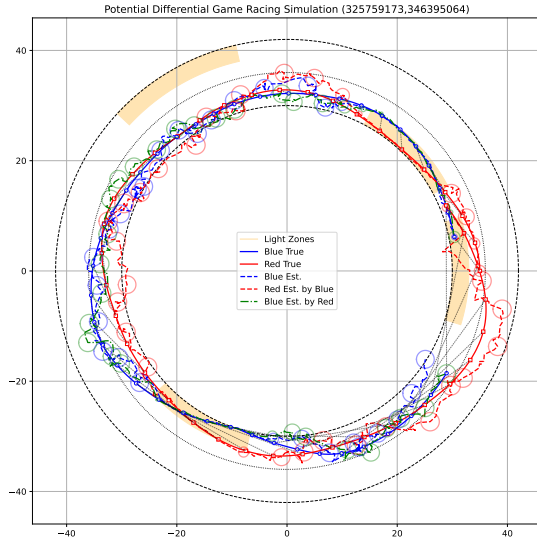


Figure 8:

Same simulation setup, but calculating Nash controls using potential Differential Game setup

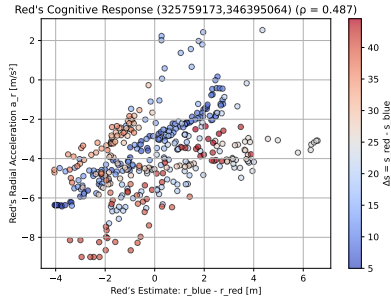


Figure 9:

Red agent's cognitive behavior

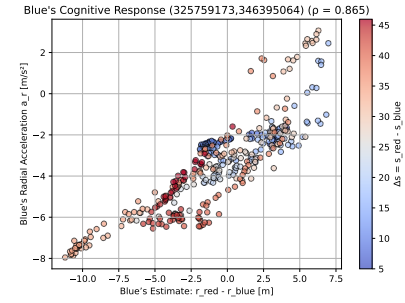


Figure 10: Blue agent's cognitive behavior

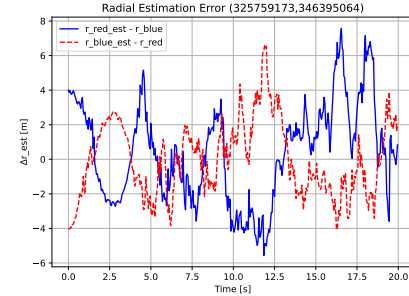


Figure 11: Radial location Estimation Error - common knowledge analysis

As we can clearly see, the results are quite the same, **but some differences are clear**. Quite good correlation performance in the analysis, and good representation of the common knowledge assumption made, but in simulation, and in other conducted simulations we did, **we've observed a more collaborative, maybe more aggressive, performance**. We can compare qualitatively the two simulations graphically, and easily observe **tighter, more aggressive and competent trajectories** of the two agents. And as mentioned, it was not exceptional, and quite consistent with repetitions we did in simulations.

6.4 Reasonable explanation - "collaborative performance"

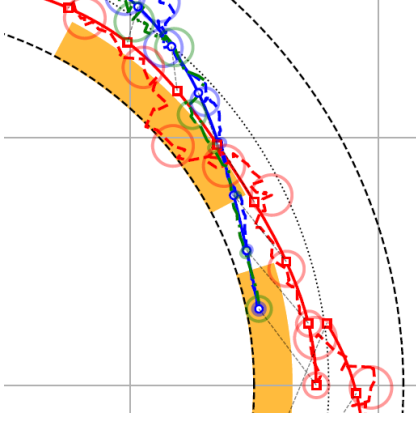


Figure 12: (SDGiBS) Starting whistle.

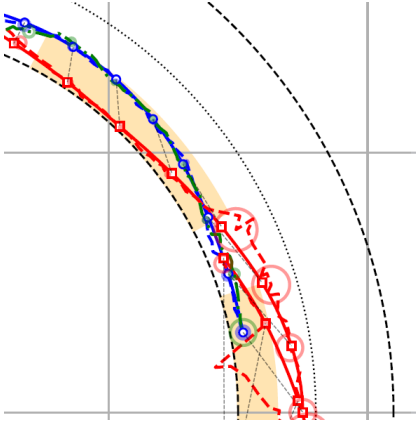


Figure 13: (PDG) Starting whistle. More aggressive turn of the red, which influences the blue turning behavior as well.

Just for the sake of clarity, we will now present the essential distinction between the two approaches to solving the optimization problem by observing main differences in simulation, and try to reason them through assumed cognitive reasoning of the agents. We must bold out that these are **Rough conclusions**, which match the analysis results from above, **and were completely consistent in all candidate simulations we conducted**. We'll notate each simulation made: (SDGiBS) - Stochastic Dynamic Games in Belief Space [1], the main papers code, and (PDG) - potential Differential Game setup, the related work paper [14]. Both simulations were conducted similarly, with the same parameters chosen to reflect the physics and penalty treatment.

Considering the most sterile scenario in race, involving the least stochasticity, is the starting whistle. We've observed a consistent

behavior where the Red agent, which started with placement disadvantage, **took the opening corner more aggressive and gridier**.

- The **key reason** for this difference lies in the **structure of the control update mechanism** and the **nature of the objective functions optimized** in both approaches.

I. Cost Decentralization (SDGiBS)

In the SDGiBS method, each agent optimizes its **own cost function** $J^i(b^i, u^i, \hat{b}^{-i})$, with dynamics modeled by belief propagation under estimation of the opponent's future trajectory.

The Nash equilibrium is reached by iterated best responses:

$$u^i \leftarrow u^i - \eta \nabla_{u^i} J^i$$

Each agent reacts to the other's policy, but optimization is local, based on individual beliefs and their gradients. As such, if red starts behind and evaluates high collision risk early on, it will moderate its speed and trajectory to reduce expected penalty, especially if blue is already occupying the inner line.

Consequence: Red behaves more conservatively in sharp turns, avoiding early inner-lane encroachment.

II. Shared Potential Cost (PDG)

In contrast, the PDG formulation assumes a **collective potential function** $\Phi(b^1, b^2, u^1, u^2)$ which approximates the sum of agents' costs under symmetry:

$$u^i \leftarrow u^i - \eta \nabla_{u^i} \Phi$$

Thus, agents **effectively collaborate to reduce global cost**, which may promote mutually beneficial strategies such as reducing joint collision risk while allowing one agent to exploit an opening.

In this scenario, red detects that aggressively cutting into the inner turn early will not significantly increase global potential (e.g., due to high uncertainty or wide separation), and that blue is unlikely to react strongly under the same policy update. As a result, red accelerates harder and turns more tightly (he manages to recognise an opportunity, **which might not be possible** using SDGiBS).

Consequence: Red **exhibits riskier** yet globally beneficial behavior earlier in the race.

6.5 Main Point

While considering which approach to implement in real life scenario, one must account for the *cognitive reasoning characteristics* which each method supports and reflects via the cost function construction. While SDGiBS might enable more noncooperative behavior, it is not guaranteeing best result for the individual agent, and considering PDG approach might benefit each and every agent through collaborating. Competing is not always about greediness, and collaboration might benefit more.

7 Conclusion and Summary

The work in [1] proposes a Belief-Space iterative LQG framework for multi-agent stochastic dynamic games, where each agent computes locally optimal feedback policies by coupling nominal trajectory optimisation with covariance evolution and anticipated opponent reactions. While the method provides a principled integration of uncertainty handling and game-theoretic reasoning, several points merit critical attention: the cubic scaling in agent count can become prohibitive for large teams, the Gaussian belief assumption may fail in multi-modal scenarios, and convergence guarantees remain local, making solution quality sensitive to initial trajectories. These aspects frame the motivation for exploring computationally lighter yet strategically rich alternatives.

To this end, two simulation studies were conducted. The first demonstrates the baseline algorithm from [1], serving both to validate our implementation and to establish a clear performance benchmark. The second introduces the Potential Dynamic Game (PDG) approach [14], which leverages the existence of a common potential function under symmetry assumptions and decoupled dynamics to reduce the search for Nash equilibrium to a **single** optimisation problem, thereby eliminating the need for multiple best-response iterations. This not only decreases computational overhead but also enables faster convergence in time-critical navigation tasks.

Results from the PDG simulation indicate that agents naturally adopt more cooperative strategies, implicitly coordinating their actions through the shared potential. This emergent collaboration led to improved collective performance, as measured by reduced collision risk and increased progress along the track for all agents, without sacrificing individual competitiveness. These findings suggest that while the original belief-space iLQG offers strong theoretical foundations, the PDG variant can achieve

comparable or superior interactive behaviours with significantly lower computational cost, marking it as a promising candidate for real-time, multi-agent autonomous navigation in uncertain environments.

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Appendix A: Symbol Table

Table 4: Main Symbols and Notation

N	number of agents
$i \in \{1, \dots, N\}$	refereed to specified agent
$\hat{\square}_k^i$	estimated @ time k for agent i
$\mathbf{x}, \mathbf{u}, \mathbf{z}$	State, control input, and measurement $\forall i, k$
$\mathbf{b}, \mathbf{s} = \begin{bmatrix} \mathbf{b}^\top, \mathbf{u}^\top \end{bmatrix}^\top$	<i>joint</i> Belief, short for belief and controls
Q^i, V^i	Action-value and value function of agent i
π^i	Optimal control policy of agent i
j_k, K_k	Feedforward and feedback gains at time k
$c_k^i(\mathbf{b}_k, \mathbf{u}_k), c_\ell^i(\mathbf{b}_\ell)$	running Cost of agent i at time k , and terminal cost
$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \mathbf{m}_k)$	State transition with process noise $\mathbf{m}_k$
$\mathbf{z}_k = h(\mathbf{x}_k, \mathbf{n}_k)$	Measurement function with meas. noise $\mathbf{n}_k$
$\mathbf{b}_{k+1} = \beta(\mathbf{b}_k, \mathbf{u}_k, \mathbf{z}_{k+1})$	Belief transition, Bayesian filtering
$\mathbf{b} = \bar{\mathbf{b}} + \delta \mathbf{b}$	Nominal + perturbation , similar for $\mathbf{u}, \mathbf{s}$
$c_{\mathbf{s},k}^i, c_{\mathbf{ss},k}^i$	Gradient and Hessian of c^i evaluated at $\bar{\mathbf{s}}_k$
$g_{\mathbf{s},k}, W_{\mathbf{s},k}$	Jacobians of g and W evaluated at $\bar{\mathbf{s}}_k$
$V_{\mathbf{b},k}, V_{\mathbf{bb},k}$	Gradient and Hessian of value at time k
$Q_{\mathbf{s},k}, Q_{\mathbf{ss},k}$	Gradient and Hessian of action-value at k
$\neg i$	all other agent except agent i